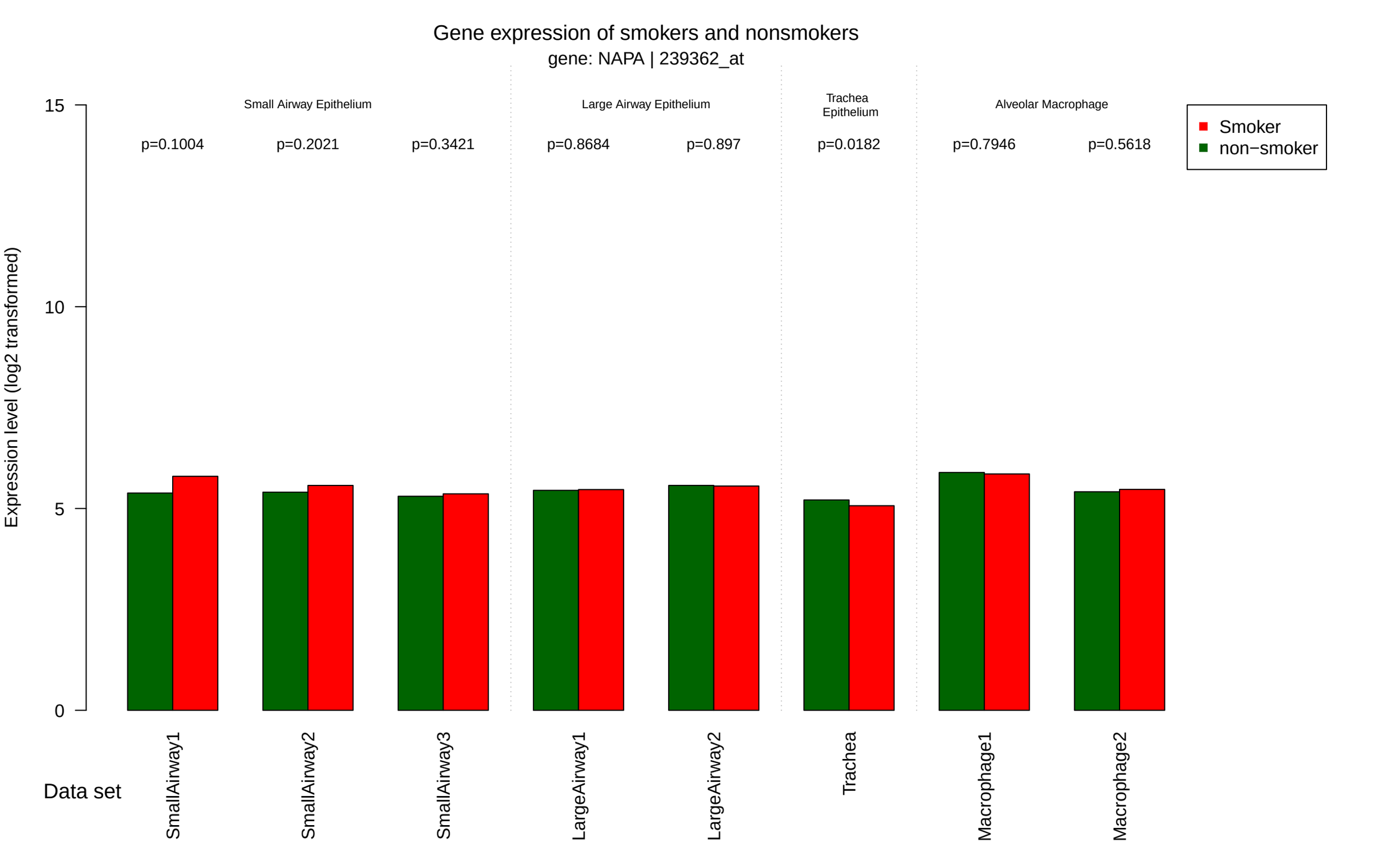
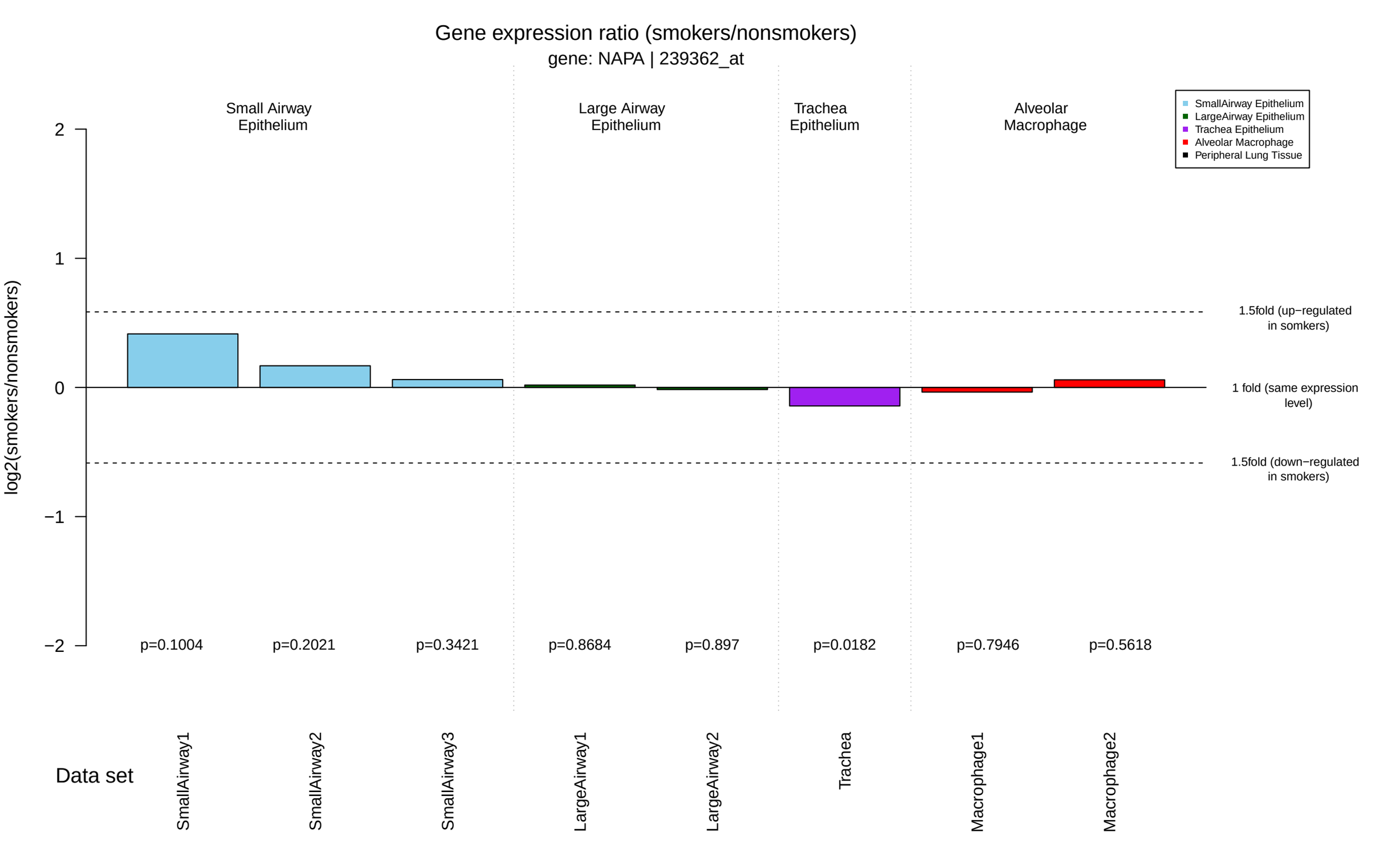
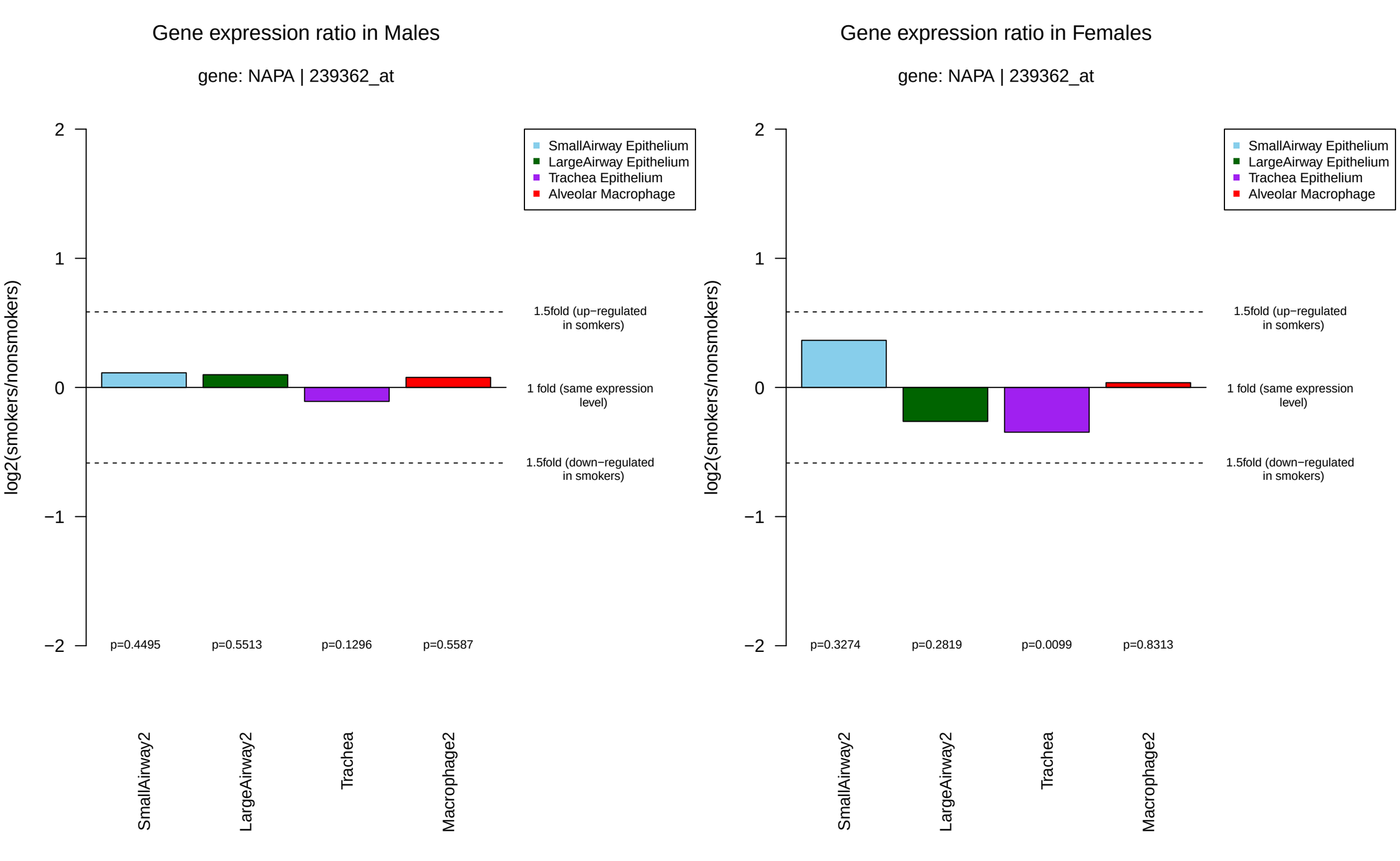
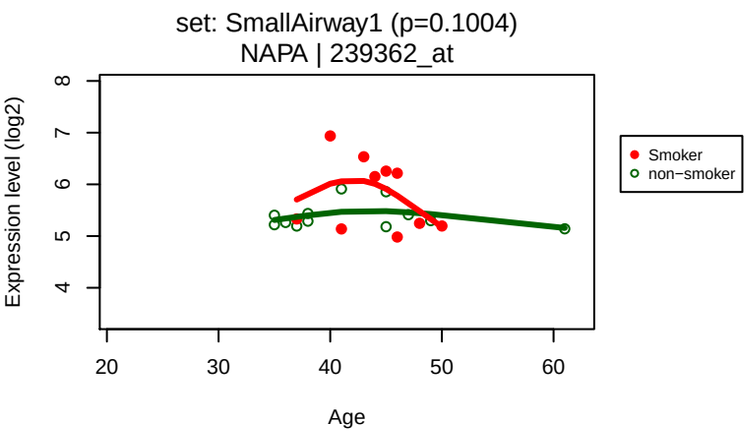
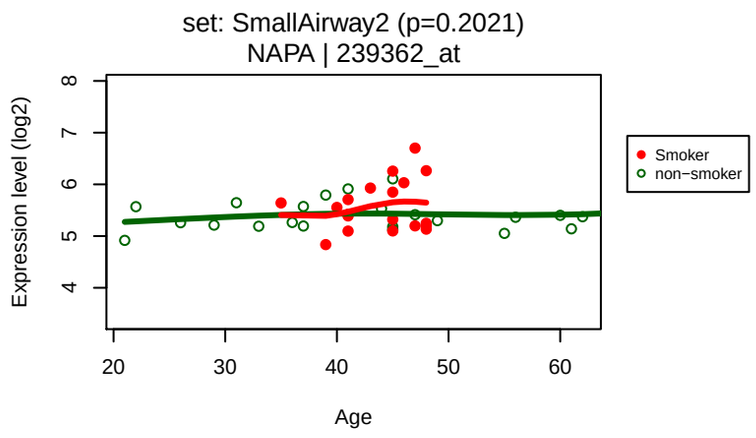
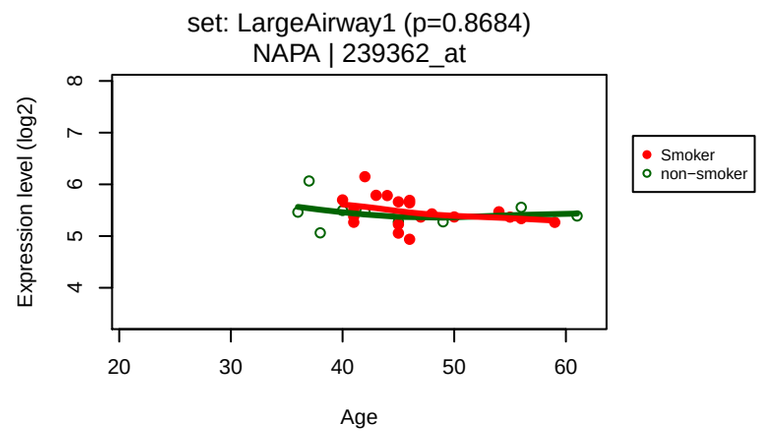

Gene expression of smokers and nonsmokers
gene: NAPA | 239362_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1004
p=0.2021
p=0.3421
p=0.8684
p=0.897
p=0.0182
p=0.7946
p=0.5618
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

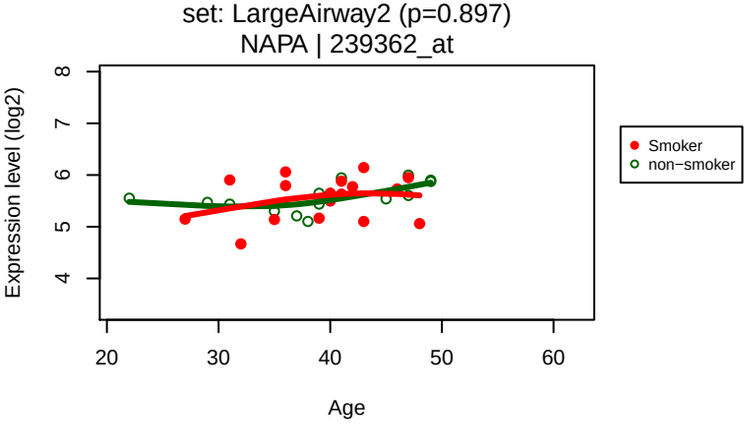

Gene expression ratio (smokers/nonsmokers)
gene: NAPA | 239362_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1004
p=0.2021
p=0.3421
p=0.8684
p=0.897
p=0.0182
p=0.7946
p=0.5618
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

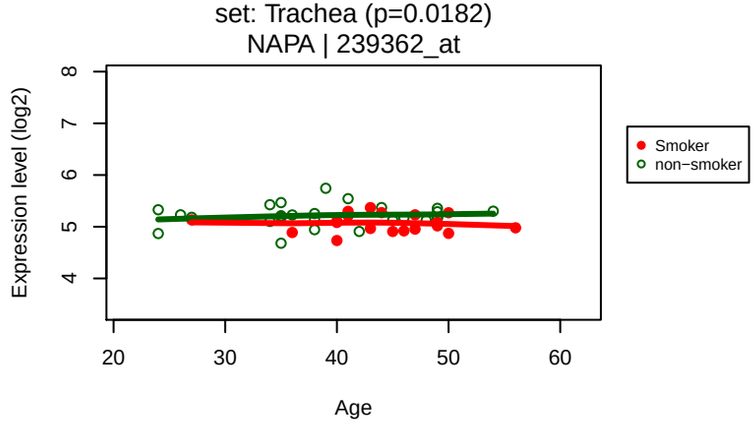

Gene expression ratio in Males
gene: NAPA | 239362_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4495
p=0.5513
p=0.1296
p=0.5587
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NAPA | 239362_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3274
p=0.2819
p=0.0099
p=0.8313
SmallAirway2
LargeAirway2
Trachea
Macrophage2

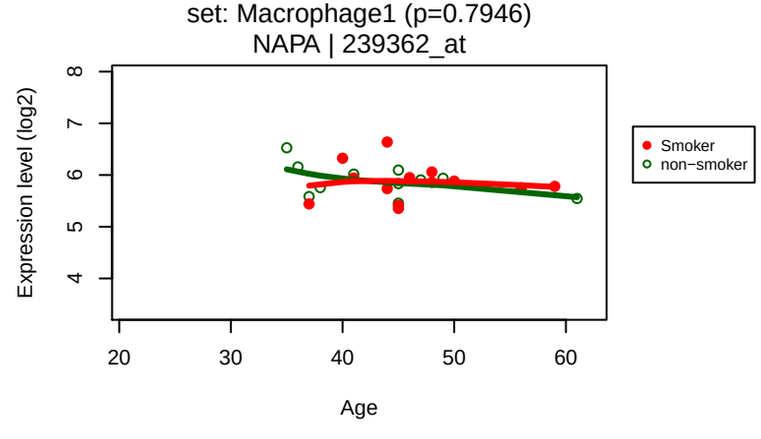

set: SmallAirway1 (p=0.1004)
NAPA | 239362_at
Expression level (log2)
Age
Smoker
non−smoker

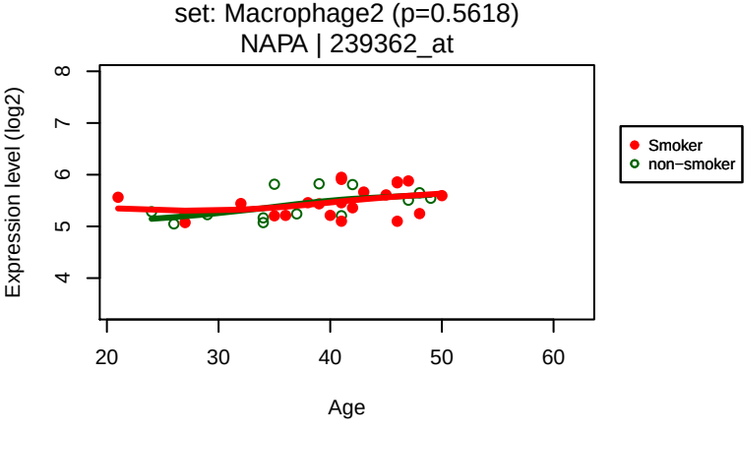

set: SmallAirway2 (p=0.2021)
NAPA | 239362_at
Expression level (log2)
Age
Smoker
non−smoker

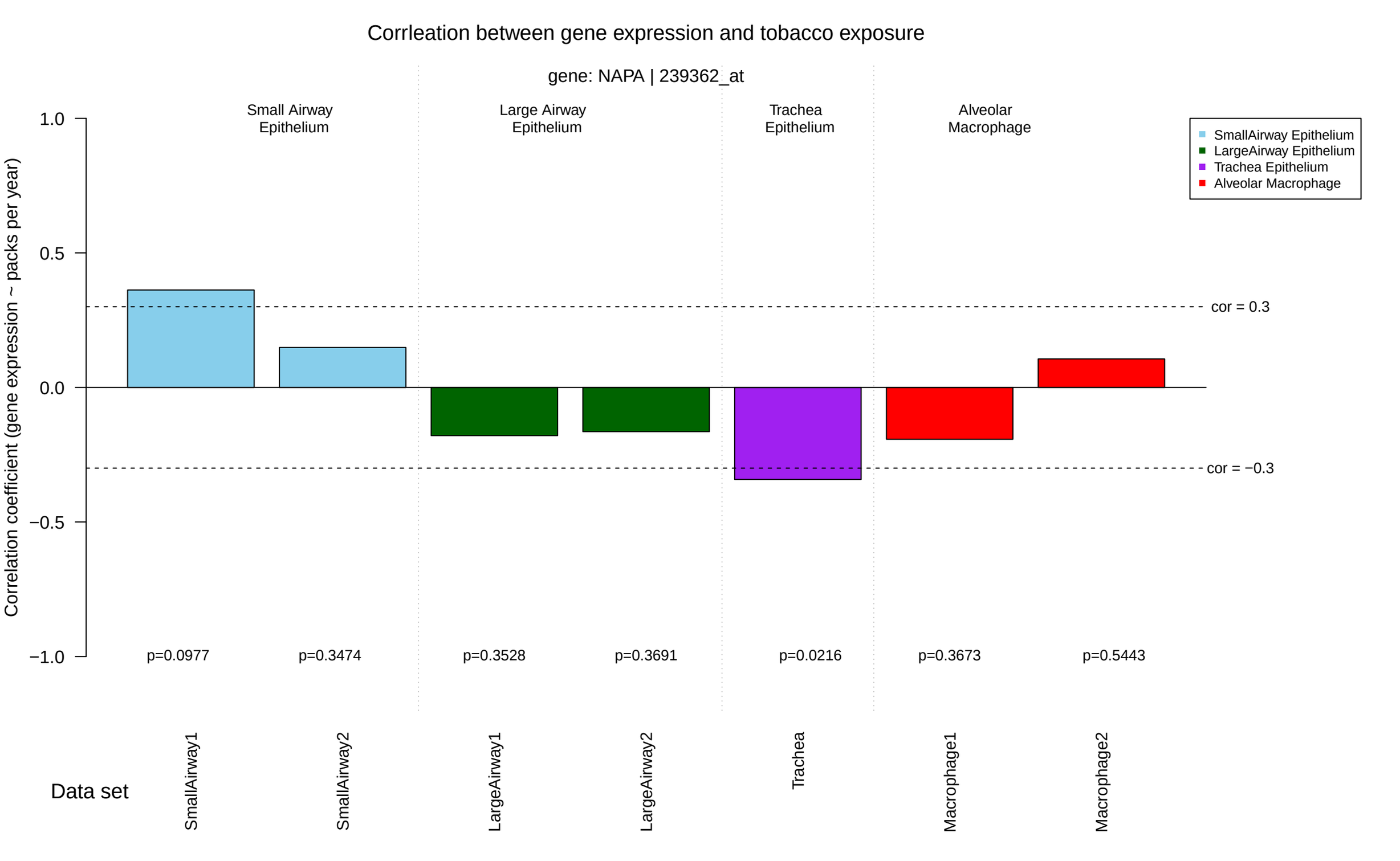

set: LargeAirway1 (p=0.8684)
NAPA | 239362_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.897)
NAPA | 239362_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0182)
NAPA | 239362_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7946)
NAPA | 239362_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5618)
NAPA | 239362_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NAPA | 239362_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0977
p=0.3474
p=0.3528
p=0.3691
p=0.0216
p=0.3673
p=0.5443
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2